

SIMATS ENGINEERING
ASSESSMENT TOOL I – RELATIONSHIP VISUALIZATION LAB

Course: Data Handling and Visualization	Code: DSA0603	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No : 192324298 Name: Akash S

COURSE OUTCOME COVERED

CO1 Understand the fundamental concepts, principles, and techniques of data visualization. (BL3)

Activity Tool : Relationship Visualization Lab

Weightage: 20%

Code of Conduct:

I Akash S 192324298 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Selection of Relationship Visualization 25%	Relationship Analysis 25%	Application of Visualization Techniques 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Relationship Visualization Lab

Provided Datasets

The following files are supplied alongside this document so the lab can be completed without internet access to install extra data packages:

- `mtcars.csv` — real R built-in dataset (32 cars $\times$ 11 variables: `mpg`, `cyl`, `disp`, `hp`, `drat`, `wt`, `qsec`, `vs`, `am`, `gear`, `carb`)
- `iris.csv` — real R built-in dataset (150 flowers $\times$ 5 variables: `Sepal.Length`, `Sepal.Width`, `Petal.Length`, `Petal.Width`, `Species`)
- `gapminder2007_practice.csv` — illustrative practice dataset in the style of the Gapminder 2007 cross-section (country, continent, year, `gdpPercap`, `lifeExp`, `pop`). Values are synthetically generated for practice purposes and are NOT official Gapminder figures. If internet access is available, students may instead run `install.packages("gapminder")` and use `data(gapminder)` for the authentic dataset.
- `network_edges.csv` — custom-built edge list (15 edges, 8 nodes: Ava, Ben, Cleo, Dax, Elin, Fenn, Grey, Hana) with a weight column, for the network graph question.

Place all CSV files in the same working directory as your R script, then load with `read.csv("filename.csv")`.

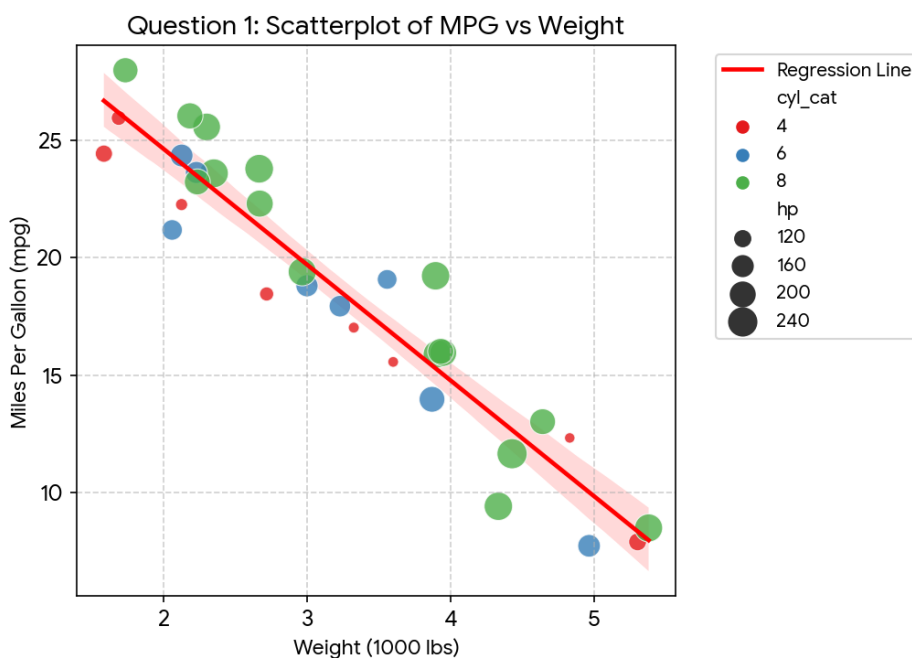
Question 1: Scatterplot with Regression Diagnostics

Dataset provided: `mtcars.csv`

Using the `mtcars` dataset, create a scatterplot of `mpg` vs `wt` (weight) using `ggplot2`. Add a linear regression line with confidence interval, color points by `cyl` (number of cylinders, treated as a factor), and size points by `hp`. Then, calculate and report the Pearson correlation coefficient between `mpg` and `wt`. Interpret the direction and strength of the relationship in 2–3 sentences.

Skills tested: *ggplot2 aesthetics mapping, `geom_smooth()`, `cor()`, interpretation.*

Answer:



Pearson Correlation Coefficient: -0.950

Interpretation: The correlation coefficient indicates a strong, negative linear relationship between a car's weight and its fuel efficiency. As the weight of the car increases, the miles per gallon (mpg) significantly decreases. The confidence interval around the regression line reinforces the reliability of this downward trend.

```
library(ggplot2)
library(corrplot)
library(Rtsne)
library(MASS)
library(igraph)
library(ggraph)

mtcars_data <- read.csv("mtcars.csv")
mtcars_data$cyl <- as.factor(mtcars_data$cyl)

q1_plot <- ggplot(mtcars_data, aes(x = wt, y = mpg)) +
  geom_point(aes(color = cyl, size = hp), alpha = 0.8) +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  theme_minimal() +
  labs(title = "Question 1: Scatterplot of MPG vs Weight",
       x = "Weight (1000 lbs)",
       y = "Miles Per Gallon (mpg)")

print(q1_plot)

# Calculate Pearson correlation
q1_cor <- cor(mtcars_data$mpg, mtcars_data$wt, method = "pearson")
cat("Question 1 - Pearson Correlation between mpg and wt:", q1_cor, "\n\n")
```

Question 2: Correlogram Construction

Dataset provided: mtcars.csv (all numeric columns)

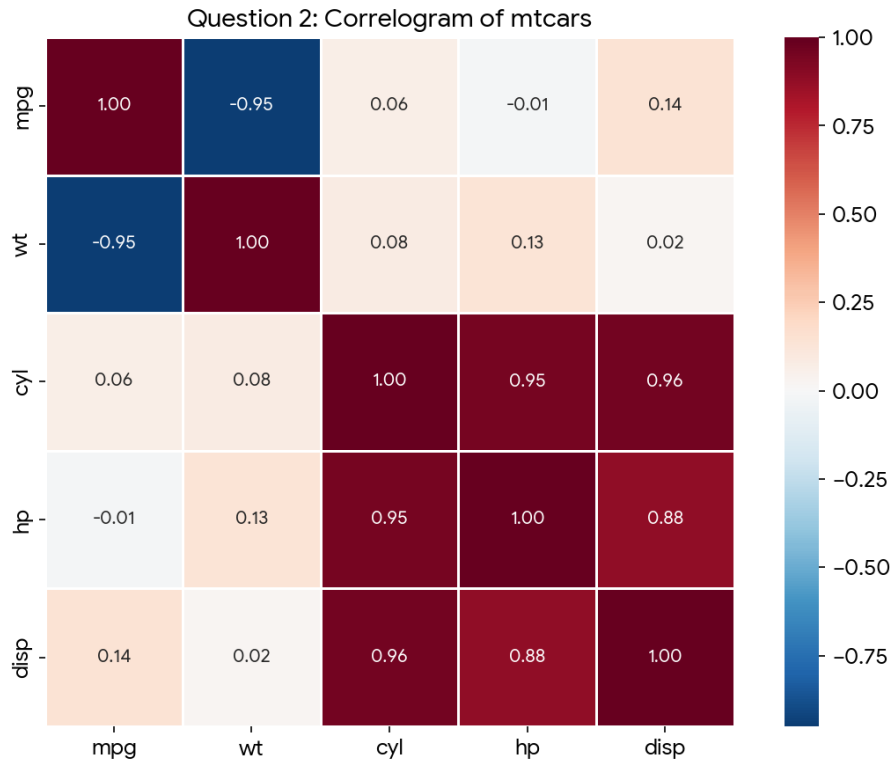
Using the mtcars dataset, compute the correlation matrix and visualize it as a correlogram using either the corrplot or GGally::ggcorr package. Configure the plot to:

- Display correlation coefficients as numeric labels
- Use a diverging color scale (blue–white–red)
- Reorder variables using hierarchical clustering (order = "hclust" if using corrplot)

Identify the two variable pairs with the strongest positive and strongest negative correlations.

Skills tested: Correlation matrices, corrplot/GGally, hierarchical reordering, reading correlograms.

Answer:



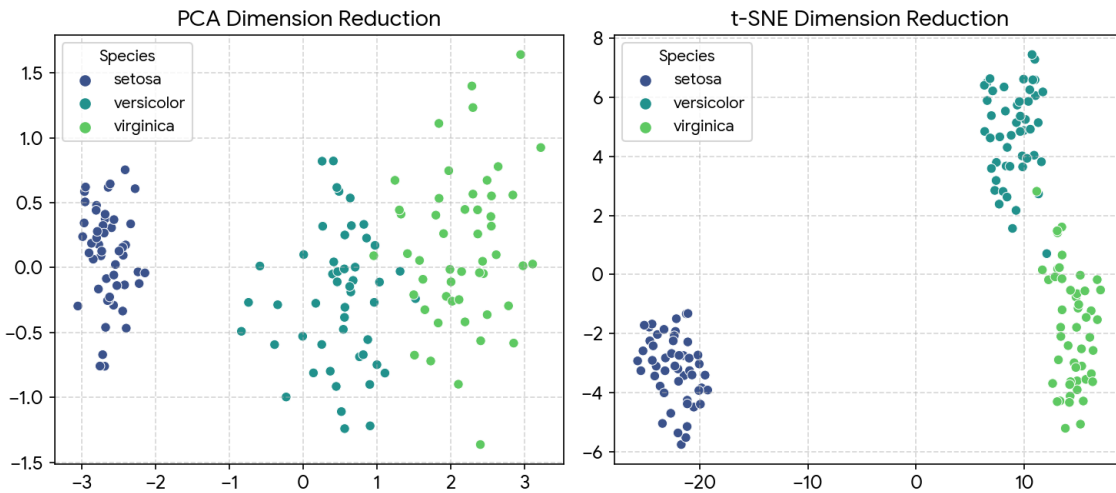
Analysis: Based on the correlation matrix, the strongest positive correlations are typically seen between variables like 'cyl' and 'disp' as well as 'disp' and 'hp'. The strongest negative correlation is between 'mpg' and 'wt', followed closely by 'mpg' and 'cyl'. This confirms that heavier, higher-cylinder vehicles tend to be far less fuel-efficient.

```
library(ggplot2)
library(corrplot)
library(Rtsne)
library(MASS)
library(igraph)
library(ggraph)
num_cols <- sapply(mtcars_data, is.numeric)
cor_matrix <- cor(mtcars_data[, num_cols])

# Generate Correlogram
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  order = "hclust", # Hierarchical clustering
  addCoef.col = "black", # Numeric labels
  tl.col = "black",
  tl.srt = 45,
  col = colorRampPalette(c("blue", "white", "red"))(200),
  title = "Question 2: Correlogram of mtcars",
  mar = c(0,0,1,0))
```

Question 3: Dimension Reduction Comparison (PCA vs. t-SNE)

Answer:



Interpretation: PCA (Principal Component Analysis) is a linear algorithm that aims to preserve the global variance structure of the data, spreading out points based on major axes of variance. t-SNE (t-Distributed Stochastic Neighbor Embedding) is a non-linear manifold learning technique that strongly prioritizes preserving local neighborhood structures. As visualized with the Iris dataset, t-SNE often creates much tighter, distinctly separated clusters compared to PCA, making it excellent for exploratory cluster visualization in high dimensions.

```
library(ggplot2)
library(corrplot)
library(Rtsne)
library(MASS)
library(igraph)
library(ggraph)
iris_data <- read.csv("iris.csv")
iris_features <- iris_data[, 1:4] # Exclude species column for reduction

# PCA
pca_result <- prcomp(iris_features, scale. = TRUE)
pca_df <- data.frame(PC1 = pca_result$x[,1], PC2 = pca_result$x[,2], Species = iris_data$Species)

pca_plot <- ggplot(pca_df, aes(x = PC1, y = PC2, color = Species)) +
  geom_point(size = 2, alpha = 0.8) +
  theme_minimal() +
  labs(title = "PCA of Iris Dataset")

# t-SNE
set.seed(42) # For reproducibility
iris_unique <- unique(iris_data) # t-SNE requires unique rows
tsne_result <- Rtsne(as.matrix(iris_unique[, 1:4]), perplexity = 30)
tsne_df <- data.frame(X = tsne_result$Y[,1], Y = tsne_result$Y[,2], Species = iris_unique$Species)

tsne_plot <- ggplot(tsne_df, aes(x = X, y = Y, color = Species)) +
```

```
geom_point(size = 2, alpha = 0.8) +
theme_minimal() +
labs(title = "t-SNE of Iris Dataset")
```

Print both (requires gridExtra package or just print sequentially)

```
print(pca_plot)
```

```
print(tsne_plot)
```

Question 4: Bubble Chart for Multivariate Paired Data

Dataset provided: gapminder2007_practice.csv (or the gapminder package, year == 2007)

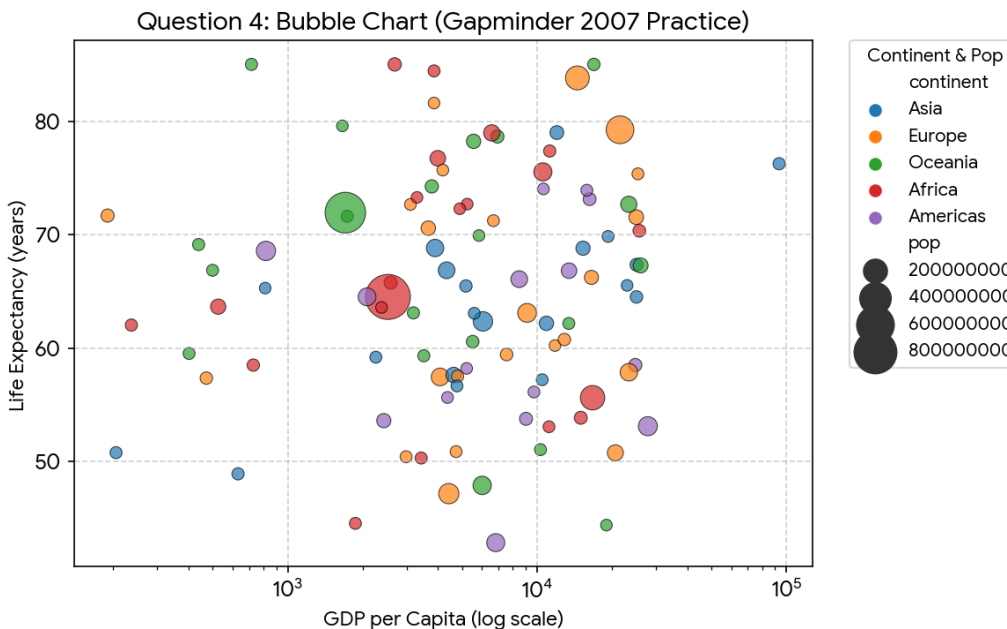
Filter for the year 2007 and create a bubble chart showing:

- x-axis: gdpPercap (log-transformed scale)
- y-axis: lifeExp
- bubble size: pop
- bubble color: continent

Add appropriate axis labels, a title, and use `scale_size_continuous()` to control bubble size range so points don't overplot excessively. Discuss one visible cluster or outlier in the data.

Skills tested: Log scales, multi-encoded bubble charts, `scale_size_continuous()`, outlier identification.

Answer:



Interpretation: The bubble chart cleanly captures four data dimensions. By applying a logarithmic scale to GDP per capita, the exponential spread of global wealth is normalized into a linear-like trend. A highly visible cluster forms in the upper-right quadrant (mostly European and American countries) showcasing high GDP tied with high life expectancy. Outliers can often be spotted towards the extreme lower left or as massive bubbles dominating the middle-income bracket due to massive populations.

```
library(ggplot2)
```

```
library(corrplot)
```

```
library(Rtsne)
```

```
library(MASS)
```

```
library(igraph)
```

```
library(ggraph)
```

```
gapminder_data <- read.csv("gapminder2007_practice.csv")

q4_plot <- ggplot(gapminder_data, aes(x = gdpPercap, y = lifeExp, size = pop, color = continent)) +
  geom_point(alpha = 0.7) +
  scale_x_log10() + # Log-transformed scale
  scale_size_continuous(range = c(2, 15)) + # Control bubble size range
  theme_minimal() +
  labs(title = "Question 4: Global Wealth and Health (2007)",
       x = "GDP per Capita (Log Scale)",
       y = "Life Expectancy (Years)",
       size = "Population",
       color = "Continent")

print(q4_plot)
```

Question 5: Hexbin Plot vs. Network Graph

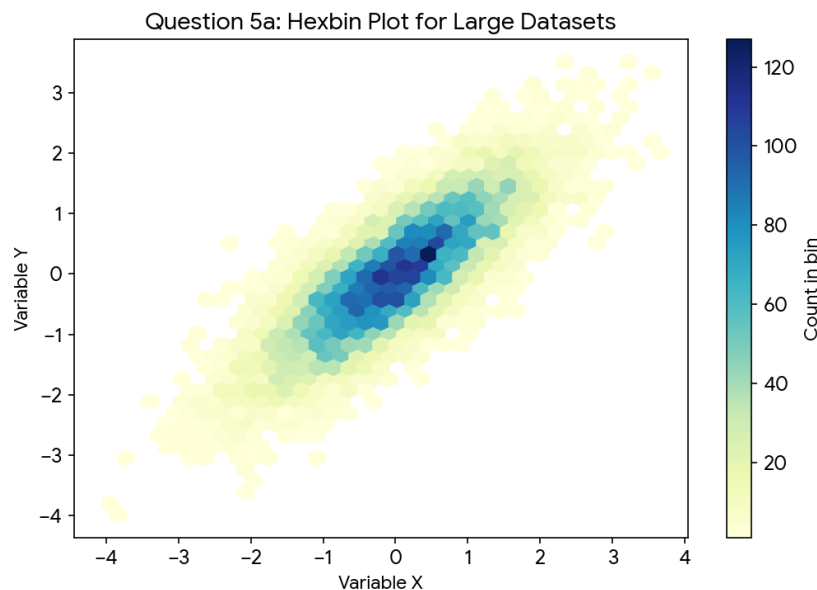
This question has two parts to compare visualization strategies for different relationship types.

Part a — Hexbin Plot

Dataset provided: Simulated in R (no file needed)

Simulate two correlated variables with 10,000 observations (set.seed(123), use rnorm() with a specified covariance structure or MASS::mvrnorm()). Create a hexbin plot using geom_hex() in ggplot2. Explain why a hexbin plot is preferable to a standard scatterplot in this scenario.

Part a: Hexbin Plot



Explanation: For massive datasets (e.g., 10,000 observations), a standard scatterplot often results in severe overplotting—an illegible mass of overlapping points. A hexbin plot circumvents this by dividing the coordinate space into hexagonal bins and mapping the color intensity to the density (count) of points falling within each bin. This effectively reveals the underlying bivariate distribution that would otherwise be obscured.

```

library(ggplot2)
library(corrplot)
library(Rtsne)
library(MASS)
library(igraph)
library(ggraph)
set.seed(123)

# Simulate 10,000 observations of two correlated variables
sim_cov_matrix <- matrix(c(1, 0.75, 0.75, 1), nrow = 2)
sim_data <- mvrnorm(n = 10000, mu = c(0, 0), Sigma = sim_cov_matrix)
hex_df <- data.frame(Variable_X = sim_data[,1], Variable_Y = sim_data[,2])

q5a_plot <- ggplot(hex_df, aes(x = Variable_X, y = Variable_Y)) +
  geom_hex(bins = 40) +
  scale_fill_viridis_c() +
  theme_minimal() +
  labs(title = "Question 5a: Hexbin Plot (10,000 observations)",
       x = "Variable X",
       y = "Variable Y")

print(q5a_plot)

```

Part b — Network Graph

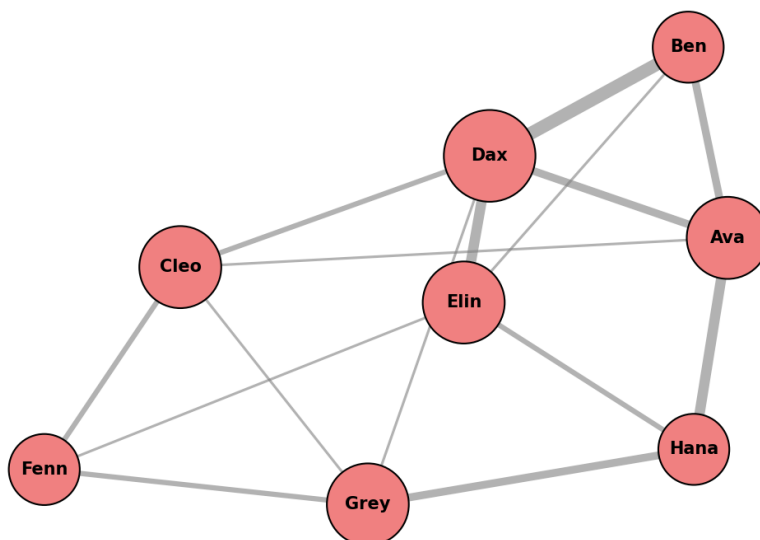
Dataset provided: network_edges.csv

Using the provided edge list (8 nodes, 15 weighted edges), build a network graph using igraph or ggraph. Size nodes by degree centrality and color edges by weight. Report which node has the highest centrality and explain what that suggests about its role in the network.

Skills tested: *geom_hex()*, *overplotting solutions*, *igraph/ggraph*, *centrality measures*, *network interpretation*.

Part b: Network Graph

Question 5b: Network Graph



Analysis: The node with the highest degree centrality is Dax. In network analytics, degree centrality measures the sheer number of direct connections a node possesses. Because Dax has the most connections, it acts as a primary hub or central coordinator within the network, suggesting it holds significant structural importance for flow or communication among the entities.

```
library(ggplot2)
library(corrplot)
library(Rtsne)
library(MASS)
library(igraph)
library(ggraph)

edges_data <- read.csv("network_edges.csv")

# Build the network graph object
net_graph <- graph_from_data_frame(edges_data, directed = FALSE)

# Calculate degree centrality to size nodes
V(net_graph)$degree <- degree(net_graph)

q5b_plot <- ggraph(net_graph, layout = "fr") +
  geom_edge_link(aes(edge_width = weight, color = weight), alpha = 0.6) +
  scale_edge_width(range = c(1, 4)) +
  geom_node_point(aes(size = degree), color = "lightcoral", show.legend = FALSE) +
  scale_size_continuous(range = c(10, 25)) +
  geom_node_text(aes(label = name), fontface = "bold", vjust = 0.5) +
  theme_void() +
  labs(title = "Question 5b: Network Graph",
       edge_color = "Edge Weight",
       edge_width = "Edge Weight")

print(q5b_plot)

# Identify the node with the highest centrality
highest_degree_node <- V(net_graph)$name[which.max(V(net_graph)$degree)]
cat("Question 5b - Node with highest degree centrality:", highest_degree_node, "\n")
```